

What is the optimal graft choice for primary anterior cruciate ligament reconstruction surgery? A living systematic review and network meta-analysis.

--Manuscript Draft--

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Corresponding Author:	Dong Woon Kim, MD Johns Hopkins University Whiting School of Engineering UNITED STATES
Order of Authors:	Dong Woon Kim, MD Jan Surma, MD Marcin Mostowy, MD, PhD Konrad Malinowski, MD, PhD
Additional Information:	
Question	Response
Select the review approach you plan to use.	Living systematic review approach
If you have selected a living systematic review approach please describe the rationale for this approach. as follow-up to "Select the review approach you plan to use."	The most recent network meta-analysis was performed in 2020, and it did not include allogenic grafts or even the third or fourth most commonly used graft types. Furthermore, outcomes reported, as well as methodological rigor and statistical validity remains to be questioned in the available network meta-analysis papers, due to the nature of difficulty in performing such analysis. Furthermore, a living systematic review, the update of which is designed to be automated only when there is a difference in statistical analysis result in one of the outcome measures, allows clinicians to always be up-to-date with the current evidence-based knowledge.
What question does your proposed review answer? (250 words)	The proposed review answers the question of whether graft choice plays a factor in determining outcomes after primary anterior cruciate ligament reconstruction surgery. And if so, which outcomes are affected by which grafts, as well as the degree of effect specific graft have on each outcome(s) that is/are found to have statistically significant differences in effect sizes. Finally, it answers whether controlling for gender, age and follow-up duration have any effect on the results of the study.
Provide a brief description of the current state of knowledge relating to the condition. Include relevant information about the intervention, test/procedure, method, prognostic factor/model you intend to review. (250 words)	The current state of knowledge in literature is that there are gold standard surgical techniques for primary anterior cruciate ligament reconstruction surgery in terms of specific grafts and suspension/fixation techniques and tools; however, as advancements in the latter over recent years has opened the door for all soft-tissue and even allogenic grafts as just as viable—if not better—alternative graft choices. Specifically, the bone-patellar tendon-bone and hamstring tendon autologous grafts have been tried-and-tested and consensus had deemed these two grafts as the 'gold standard' choice for orthopedic surgeons. However, the quadriceps tendon—particularly without the bony attachment—was found to produce non-inferior results compared to those with the patellar bone attachment. This finding opens the door for comparing all soft-tissue grafts with other that share this similar characteristic, such as the hamstring and peroneus longus tendon autologous grafts. As there are

	numerous studies comparing the various grafts, but not one comprehensive source from which to guide evidence-based practice, this review paper seeks to add to fill this gap in current knowledge by comparing the available randomized controlled trials to produce an evidence level I network meta-analysis to guide clinicians in their practice.
Why is this an important question? Is it especially topical at present? If so, please explain why. Summarise the importance of the question to patients, health professionals, communities or policy makers and the degree of uncertainty in the existing evidence base. (250 words)	This is an important question because total knee arthroplasty (TKA), or total knee replacement is one of the most commonly performed surgical operations in the world. However, studies have found a sole indication for such a procedure, and that is osteoarthritic knee pain. Pain that is chronic, and strong in intensity that it permanently degrades the indicated patients' quality of lives. TKA, however, also limits patients' locomotive capabilities, and is not a procedure that is synonymous with restorative. The most commonly cited research topic in orthopedic surgery by orthopedics surgeons was anterior cruciate ligament reconstruction (ACLR) surgery, with the link between this and TKA being that the former (ACLR) is the most common cause of non degenerative and thus, the most common modifiable risk factor for preventing TKA and osteoarthritis. If we are able to answer certain questions through research, then we can advance our collective body of knowledge surrounding one of the most debilitating musculoskeletal conditions and preventing those from developing osteoarthritis after ACLR.
Has the review been funded or commissioned, and if so, by whom? If your review has been commissioned by Cochrane, please note: "Commissioned by Roses Parker, Cochrane Commissioning Editor".	No.
Word count From July 1, 2026, new submissions to Cochrane must comply with the below word limits: - Reviews and updates will have a maximum of 12,000 words. - Protocols will have a limit of 6,000 words. Please confirm you've read the above and are aware of these word limits.	I/We confirm
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Yes

Will the review address issues of health equality (targeting disadvantaged populations, reducing social gradients, etc). If so, please explain why?

No

Proposed review eligibility criteria

For an intervention review (or overviews of intervention reviews), specify the population, intervention, comparators and outcomes (PICO) for your review. Refer to

Inclusion criteria:
(P) Adult patients .
(I) Primary anterior cruciate ligament reconstruction surgery.
(C) Bone-patellar tendon-bone, hamstring tendon, quadriceps, or peroneus longus tendon autologous grafts, or Achilles tendon or tibialis anterior/posterior tendon allogenic grafts.

the Cochrane Handbook .	(O) Primary outcomes of interest: IKDC subjective, Lysholm, Tegner, pivot shift, Lachman, instrumental laxity, graft rupture, graft failure, revision rate (S) Randomized controlled trials. Exclusion criteria: (P) Pediatric patients . (I) Revision anterior cruciate ligament reconstruction surgery, anterior cruciate ligament repair. (C) Bridge-enhanced augmented repair (BEAR), lateral extra-articular tenodesis (LET). (O) Secondary outcomes of interest Marx, KOOS, (S) Observational studies
Participants/population (150 words) Outline the types of populations to be included and excluded. Consider any pertinent characteristics that will be considered eligible or ineligible (e.g. demographic factors such as age and gender, the type/stage of disease/condition, medication at baseline, co-morbidities, health issues, setting).	Adult patients with post-traumatic grade III (complete tear) of unilateral anterior cruciate ligament are included. Both genders are included, and upper limit in age is unrestricted. Only studies with mean age greater than or equal to 18 years old across all patients will be presumed to have targeted a population that is considered skeletally-mature.
Intervention(s) (150 words) Outline the details of the intervention you wish to investigate. Consider the dose, intensity, mode of delivery, and combinations of interventions. Are there variations you wish to exclude?	Primary anterior cruciate ligament reconstruction (ACLR) surgeries are included. ACLRs performed simultaneously on both knees, in conjunction with other ligamental reconstructions aside from meniscal, and revision ACLRs were excluded.
Comparator(s) (150 words) Outline the details of the comparator(s) you wish to investigate.	Autologous grafts: bone-patellar tendon-bone, hamstring tendon, quadriceps tendon, and peroneus longus tendon are included. Allogenic grafts: Achilles tendon and tibialis anterior and posterior are also included. All other graft types, such as synthetic grafts and allogenic versions of commonly used autologous ones are excluded (the Achilles and tibialis tendon grafts are included despite being allogenic because autologous versions cannot be grafted in patients due to the important roles the muscle from which the tendon is attached plays in normal physiologic biomechanics).
Outcome(s) (150 words) Outline the details of critical and important outcomes you wish to investigate. Where established Core Outcomes have been developed for your review question, they should be included amongst the list of outcomes unless there is good reason to do otherwise. Consult the COMET Initiative website at www.comet-initiative.org to identify whether relevant Core Outcomes have been established. Please reference any known core outcome sets.	The outcomes of interest were measures of (1) patient-reported outcome measures (PROMs) of patient satisfaction as surrogate measure for subjective quality of life—answering the question of how much is the patient able to live as close to pre injury state; (2) Objective measures of translational and rotational knee stability; and (3) Adverse events and complications. In (1), the primary (most commonly reported) outcomes were included: IKDC subjective knee form (SKF), Lysholm and Tegner. For (2), instrumental laxity (side-to-side difference in millimeters) of anteroposterior translation by the anterior drawer test, pivot shift and Lachman tests were included. For (3) graft failure, including re-re-rupture of graft (confirmed via imaging), clinical failure (defined by clinical failure during objective measures) and revision surgeries (arthroscopic confirmation of tears). Other outcomes that were excluded were due to their deemed incompatibility for comparisons across all grafts. For example, hamstring and quadriceps muscle strength recovery is not a relevant outcome measure for determining effectiveness of ACLR when comparing across grafts in which some were not taken from the patients'

	hamstring or quadriceps muscle, or is a component that disrupts such biomechanical physiology. Donor site morbidity is another example of an outcome that may be out of range in scope, due to the variety of definitions for consistent applicability in comparisons. Other outcomes such as Marx, KOOS, VAS, were excluded due to the low number of studies that reported these outcome measures, in comparison to the other ones that surrogatively measure the same thing.
Study type (150 words) Outline the types of studies to be included. Please describe specific features of studies which would lead to their inclusion rather than simply study labels (e.g. description of the unit of allocation where necessary).	Comparisons. Other outcomes such as Marx, KOOS, VAS, were excluded due to the low number of studies that reported these outcome measures, in comparison to the other ones that surrogatively measure the same thing.
Reviews addressing questions other than effects of interventions For proposals relating to diagnostic test accuracy, prognosis, methodology or qualitative synthesis questions, please use the free text box below to describe the concepts that will be used as the basis for the eligibility criteria.	
Will the review generate substantial public interest? If so, please explain why?	Yes, this is the most common sports-related injury that requires surgical intervention. The public's knowledge of ACL injuries in professional athletes and their detrimental effect carries ample weight in this regard. Furthermore, it is the most commonly cited topic with the most number of questions asked in certification exams for orthopedic surgeons entering the field, and lastly, it is the most common modifiable risk factor for one of the most common surgeries performed worldwide.
Does this proposed review overlap with an existing Cochrane or non-Cochrane review or protocol? Cochrane will only proceed with new proposals that do not overlap with existing Cochrane protocols or reviews, or where there is suitable rationale for overlap with an existing Cochrane protocol or review. Please check the Cochrane Library for overlap with published reviews and protocols.	No, it does not overlap
Evidence available: Is there RCT evidence available in this area? If not, please provide information about the types of studies available.	Yes there are RCTs available in this area; however, the exact number is unknown, but enough studies for a network meta-analysis is thought to be probable.
Size of proposed review: Do you think that more than 50 studies will be included in the review?	Yes
If yes, what is the justification for one large review, rather than splitting the topic	There will most likely be close to 50 RCTs, across the six subgroups included in the network meta-analysis.

into several smaller reviews?

as follow-up to "Size of proposed review: Do you think that more than 50 studies will be included in the review?"

A network meta-analysis requires that all studies be included in the same review. A network meta-analysis of RCTs and observational studies would be too large, indeed. Therefore, several smaller reviews of single-arm results will be produced to complement the network meta-analysis of two- and multi-arm RCT studies. The results from both will be compared via statistical analysis

Are you aware of any ongoing studies in the field? How many would you expect to complete in the next 24 months?

as follow-up to "Size of proposed review: Do you think that more than 50 studies will be included in the review?"

There are no on-going studies on the topic. The network meta-analysis is expected to take several months, and the accompanying meta-analysis papers after its completion will utilize the templates created and thus are expected to be completed in a much shorter time-frame. Nevertheless, the project timeline is set so that all seven papers are expected to be completed and submitted before the end of the calendar year. The following year would follow a similar pattern as that of the papers after the network meta-analysis paper for a total of over 10 papers in the next 24 months, in a conservative estimation.

Information Specialist expertise

Yes - within the author team

Cochrane reviews must be prepared by at least two people, and often require more than two authors. A team should have a range of skills and experience including clinical topic expertise and methodological knowledge. See Cochrane Handbook for Systematic Reviews of Interventions (<https://training.cochrane.org/handbook/current/chapter-ii#section-ii-2-1> and <https://training.cochrane.org/handbook/current/chapter-ii#section-ii-2-2>)

Some Cochrane groups are able to provide specific support; for example, with searching medical databases. However, many author teams work independently. To help us assess what support you might need to develop your review, please answer the following question:

Do you have access to an Information Specialist who will assist you with developing and running literature searches of medical databases?

Important: Please note that approving a proposal does not guarantee that a protocol will be accepted. You will be given a time limited period to prepare a draft protocol for submission. If you do not submit a draft for editorial consideration by the end of that period, Cochrane will reserve the right to assign your title to another team.

Yes

Please confirm you have read the above by selecting 'Yes'.

Author Comments:

An update to the list of authors was requested, and since there is no specific place to append this information, it is being written here:

1. Dong Woon Kim, M.D.
2. Jan Surma, M.D.
3. Shayden Bernas
4. Marcin Mostowy, M.D., Ph.D.
5. Marko Puszak, M.D.
6. Konrad Malinowski, M.D., Ph.D.